

SAGE-SRL

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I. INTRODUCTION

The rapid proliferation of Large Language Models (LLMs) has reshaped the landscape of intelligent tutoring systems (ITS). Unlike earlier rule-based or scripted tutors, models such as ChatGPT can explain concepts, debug code, and synthesize examples in real time, promising scalable and personalized instruction for millions of learners [1]. But personalization at this scale only works if the tutor can correctly judge *what each learner has actually mastered*. That judgment is the decision on which everything else depends: what to review, what to teach next, and what to unlock all follow from it [2]. A tutor who misjudges mastery does more than give imperfect advice, it moves students past concepts they have not learned, and the gap only widens as the material builds on itself.

For decades, tutoring systems have judged mastery with Bayesian Knowledge Tracing, which treats each concept as a single hidden skill, learned or not, and revises that estimate from one stream of evidence—usually quiz answers [3]. Later work made the idea more powerful: one line replaced the simple statistical model with a neural network [4], and another learned to follow many skills at once [5]. Useful as these are, they share a blind spot. Each still reduces a student’s grasp of a concept to a single number, and that number says nothing about *what kind* of understanding the evidence actually showed. A different family of models, borrowed from educational testing, does split a skill into several parts [6], [7], but those are built to score a finished exam, not to keep up with a student as a lesson unfolds.

This blind spot leads to a specific and costly mistake. Learning a concept is rarely a single skill. Take any topic in introductory programming: a student might recall what a loop does, write a working loop, and reason about why one misbehaves—three distinct abilities that a single score quietly blends into one. A student who only ever answers recall questions can push that one number to the top and be marked as having mastered the topic, without ever writing or analyzing a line of code. We call this the **Evidence Diversity Problem**: a tutor certifies mastery from just one kind of evidence and credits a student with skills they have never shown. The trap is easy to fall into, because some evidence is far cheaper to gather than the rest—quiz questions are quick to ask and grade automatically, while writing and analyzing

code is not so; a tutor left to its own devices leans on the easy evidence and overlooks the rest.

The cost is highest when an instructor, rather than the student, decides what comes next. Our earlier system, SAGE, was built for exactly this setting [8]. It lets instructors enforce the shape of a course which topics are in scope, and which ones a student must clear before moving on—by checking each student’s progress against the set of topics the system treats as mastered. SAGE settled the question of *control*: it gives instructors a dependable, auditable say over where each student can go. What it never settled was the question of *evidence*. SAGE keeps a record of mastered topics, but it has no real way to decide what belongs there; a topic is marked as mastered when the student completes it or an instructor says so, not because the system has weighed any evidence that learning happened. The gate is therefore only as trustworthy as a record taken on faith and the Evidence Diversity Problem fills that record with topics a student has met from a single angle. The SAGE paper itself named this gap, pointing to a genuine, evidence-based model of student learning as the main thing left to build.

This paper builds that missing piece and joins it to SAGE. Instead of tracking one score per concept, we track three one for each kind of understanding: recalling an idea, putting it into practice, and reasoning about it. These three levels are not arbitrary: they follow Bloom’s widely used hierarchy of learning objectives [9], and they line up with the kinds of evidence a tutor can readily gather—a question answered, a task carried out, a piece of work explained. In introductory programming, the domain in which we test the model, they map cleanly onto quizzes, coding exercises, and code-review tasks. Each piece of evidence updates only the score it belongs to, and a concept counts as mastered only when all three scores are high at the same time. That judgment is what fills SAGE’s record of mastered topics, so the two halves fit together cleanly: our model decides what a student has genuinely earned, and SAGE’s existing machinery enforces the course structure on top of it. This mirrors a familiar division between the structure an instructor imposes (instructor-regulated learning) and the monitoring of a learner’s own progress that self-regulated learning calls for, work this system performs on the student’s behalf. SAGE provided the first; this paper adds the second, and the second is what lets the first rest on earned rather than assumed mastery. Each piece of this has a precedent. Models from educational testing can require a student to



Fig. 1: The Evidence Diversity Problem. Three students with different competence profiles across recall, code-writing, and code-reasoning (top) receive near-identical scores from a single-estimate model (bottom), which certifies A and B as masters of competencies they never demonstrated.

show several distinct skills before crediting mastery [10], [11], and knowledge tracing has long revised its estimates as a student works. What we have not seen brought together is the whole arrangement: each kind of evidence feeds its own running estimate, those estimates update live during the session, and a concept stays locked until all of them are met so mastery cannot be reached without evidence of every kind. Two questions shape the design.

First, how do we produce a mastery signal an instructor can trust? Every time a student does something—answers a question, writes a snippet, explains a bug—we label the kind of skill it demonstrates and use it to update only the matching score. A concept is marked as mastered, and added to the record SAGE relies on, only once all three scores clear their mark. The course gate now rests on evidence the system has actually weighed rather than on a status taken for granted.

Second, how do we stop one kind of evidence from standing in for the rest? Sorting evidence by type is not enough on its own; the rule for declaring mastery has to demand breadth. Because each piece of evidence moves only its own score, no amount of quiz-answering can raise the score for writing or for reasoning—and the certification rule asks for more than a high score: each kind of understanding must be demonstrated correctly a minimum number of times. This gives a guarantee we state and prove later (Theorem 2): the system can mark a concept as mastered only if it has observed at least $n_{\min}$ correct demonstrations of every kind of evidence. A student who has only answered recall questions, however many, can never be certified—not because we penalize them, but because the other evidence simply is not there. We further bound the odds of certifying by luck:

a student guessing at every kind of task falsely certifies a concept with probability below one in a billion (Corollary 3). Mastery from a single kind of evidence is ruled out by the design itself; mastery by guessing is ruled out to within a quantified, negligible probability.

To test the combined system we [PLACEHOLDER: describe the study—for example, simulations over synthetic learners and/or a pilot in an introductory programming course]. We measure [PLACEHOLDER: for example, how often the system would have certified mastery from one kind of evidence alone, how closely its mastery judgments track later performance, and how often students are advanced past material they had not earned], and find that [PLACEHOLDER: headline result]. One property holds no matter the data: because mastery requires every kind of evidence, the system never certifies a concept from a single source—and the experiments confirm it.

Our contributions are as follows:

- We name and define the **Evidence Diversity Problem**: the systematic over-crediting that arises when a tutor judges mastery from a single kind of evidence. We show why it is especially harmful when an instructor governs progression, because an inflated record of mastery quietly corrupts the prerequisites the course depends on.
- We introduce a student model that follows each concept across three kinds of understanding rather than one—a **factored, multi-skill knowledge-tracing model**. Each kind of evidence updates only its own estimate, the estimates update live as the student works, and a concept is declared mastered only when all three are met. This combination—routing evidence by type, updating

online, and gating on an all-or-nothing mastery rule—is the paper’s main technical contribution, and it produces a record of mastered topics that an instructor-regulated tutor can use directly.

- We prove an **Evidence Diversity Guarantee**: a concept cannot be marked as mastered unless the system has observed at least $n_{\min}$ correct pieces of evidence of every required kind—a property of the architecture itself, independent of the calibrated parameter values, that makes mastery from a single kind of evidence impossible by construction rather than merely unlikely. We complement the guarantee with quantitative bounds on false certification: a non-master guessing at tier k certifies that tier with probability at most $(p_G^{(k)})^{n_{\min}}$ over a fixed battery, and at most 10^{-9} for a full concept.
- We join this student model to our earlier instructor-regulated system, SAGE [8], so that its prerequisite gate finally acts on earned rather than asserted mastery, together with the supporting parts that make a tutor usable in practice—review scheduling, adapting how material is taught, tracking misconceptions, and watching for frustration—and [PLACEHOLDER: state the evaluation result matching the study above].

II. RELATED WORK

A. Validity and the Evidence Behind a Mastery Decision

Whether a mastery judgment can be trusted is, at bottom, a question of measurement validity. A long tradition in educational measurement holds that a claim about a competency is only as sound as the evidence assembled for it, and warns specifically against *construct underrepresentation*—treating a narrow slice of evidence as if it certified a broad ability [?]. Evidence-centred design makes this reasoning explicit, casting assessment as an argument that runs from a student’s observed work to claims about what they know [?], and the mastery-learning tradition likewise insists that advancement wait on demonstrated competence rather than mere exposure [?]. Seen through this lens, the Evidence Diversity Problem is a concrete instance of construct underrepresentation: when a tutor certifies mastery of a many-sided concept from a single kind of evidence, the inference outruns its support. The rest of this section asks how existing learner models fare against this standard.

B. Knowledge Tracing and the Single Mastery Estimate

The standard way a tutoring system decides what a student has learned is knowledge tracing. Bayesian Knowledge Tracing treats each concept as a hidden skill that is either learned or not, and updates the probability of that skill from the student’s record of correct and incorrect answers [3]. Later models widened the approach without changing this premise: Deep Knowledge Tracing replaced the simple statistical model with a recurrent network [4], memory-based models began following many skills at once [5], and attention-based models improved how past practice is weighted when

predicting the next response [?]. What makes knowledge tracing valuable for tutoring, rather than for testing, is exactly its online character: it revises its estimate after every interaction, so a tutor can act on it in the middle of a lesson instead of waiting for a final exam. Recent tutoring systems have paired knowledge tracing with retrieval to tailor responses to a predicted learner state [12], but the tracked state is still a single per-skill estimate, and the decisions built on it remain entirely on the system’s side. Across this line of work, however, a learner’s grasp of a concept is summarized as a single number, and that number is driven by one undifferentiated stream of correctness. The form of the evidence—whether the student recalled a fact, carried out a procedure, or reasoned about a result—is not represented, so a model has no way to notice when its estimate rests on only one kind of task. This is precisely the setting in which the Evidence Diversity Problem arises: heterogeneous evidence is pooled into one score, and a high score earned from a single source is indistinguishable from broad competence.

C. Multidimensional and Cognitive Diagnosis Models

A separate tradition, rooted in educational measurement, does represent more than one ability per concept. Cognitive diagnosis models infer which of several underlying skills a student possesses from their test responses [6], and multidimensional item response theory places a learner on several ability scales at once [7]. Some of these models are explicitly conjunctive: the DINA family, for instance, treats a correct response as requiring *all* of the skills an item demands, so a student is not credited unless every required attribute is present [10], [11], and later frameworks generalized this to a broad family of diagnostic models [?]. Our work shares the conjunctive intuition with these models, and we do not claim that idea as new. The difference lies in purpose and mechanism. These models are primarily designed for assessment and attribute diagnosis rather than for continuous tutoring decisions: their structure is fixed by expert design, and the attributes are tied to the items of a particular test. Dynamic and longitudinal variants of cognitive diagnosis do exist, but they are generally not used as learner models that route heterogeneous evidence by cognitive type and maintain online mastery estimates for instructional control. As a result, these models do not, on their own, yield the live, evidence-grounded record of mastered topics that a tutor needs to decide what to unlock next.

The choice of which dimensions to track is itself grounded in a long-standing framework. Bloom’s taxonomy organizes educational objectives by cognitive process—remembering, applying, analyzing, and so on—and its later revision refined these categories [?], [9]. The taxonomy is widely used in curriculum design and assessment, including in computing education, to ensure that instruction and testing reach beyond rote recall. These frameworks, however, are descriptive: they classify objectives and items, but do not provide a learner

model that maintains and updates mastery estimates across cognitive levels during instruction. Our three dimensions—recall, procedural, and applied—are a deliberately compact, Bloom-aligned decomposition, chosen so that each level corresponds to a distinct and separately observable kind of evidence a tutor can collect.

D. Regulated Learning and Access-Controlled Tutoring

A parallel concern is who controls a student’s path through the material. Self-regulated learning describes how a student directs their own study across cycles of planning, performing, and reflecting [13], [14], and a recurring finding is that these regulatory skills must be actively scaffolded rather than assumed, with responsibility shifted to the learner as competence grows [15]. Instructor-regulated learning, by contrast, concerns the boundaries an instructor imposes—which topics are in scope and in what order they may be taken. Tutoring systems, including LLM-based ones, typically keep these regulatory decisions on the system’s side rather than the learner’s [1], [16]. Our prior system, SAGE, addressed the instructor’s side of this picture: it grounds responses in course materials and enforces curriculum boundaries and prerequisites through a three-layer access-control gatekeeper—reaching 90% security and 80% curriculum compliance—and checks each student’s progress against the set of topics it treats as mastered [8]. SAGE thus provides dependable access control over a set of mastered topics, but no principled mechanism for inferring what belongs in that set: membership is conferred by completion or instructor status rather than earned through evidence of learning. The present work supplies exactly that missing mechanism and feeds its output directly into SAGE’s existing gate.

A clarification on terminology is in order. In the literature, self-regulated learning refers to a student’s own metacognitive activity—setting goals, monitoring comprehension, and evaluating their progress [13], [14]. Our system does not set out to build these metacognitive skills in students. Instead, it performs much of this regulation on the learner’s behalf: the learner model monitors mastery, a misconception check flags confident errors, and review is scheduled automatically, while the student contributes a small number of genuine metacognitive signals, such as goal statements and confidence ratings. Where we speak of self-regulated learning, then, we mean this system-supported regulation rather than autonomous metacognition—a distinction we revisit when discussing limitations.

E. Summary

In short, the two capabilities the Evidence Diversity Problem demands are each present in prior work, but never together. Knowledge tracing updates a mastery estimate online as a student works, but collapses every kind of evidence into one score. Cognitive diagnosis models separate competence into several dimensions and can require all of them at once, but are designed for after-the-fact diagnosis rather than tutoring control, and do not route evidence by

TABLE I: Capabilities of representative learner-modeling and tutoring approaches. ✓: supported; △: supported only by some variants; ×: not supported.

Capability	BKT	DKT	CDM	SAGE	Ours
Online updating	✓	✓	△	×	✓
Multi-dimensional mastery	×	×	✓	×	✓
Evidence-type routing	×	×	×	×	✓
Conjunctive mastery	×	×	✓	×	✓
Curriculum enforcement	×	×	×	✓	✓

type. And instructor-regulated tutors such as SAGE can act on a mastered set but cannot themselves determine what belongs in it. Table I summarizes these gaps; the evidence-type-routing row, supported by no prior approach, is the one our model is built around. The contribution of this paper is a learner model that brings these strands together—routing each kind of evidence to its own estimate, updating those estimates online, and granting mastery only when all are met—so that the resulting record of mastered topics can be trusted by an instructor-regulated tutor. In effect, the system carries out the learning-state monitoring that self-regulation would otherwise demand of the student, and it is that monitoring which finally lets the instructor’s regulation rest on earned rather than assumed mastery.

III. SYSTEM OVERVIEW

Before formalizing the learner model, we describe how it sits inside a complete, deployed tutor. The system is organized into two parallel tracks that meet at a security gate (Fig. 2). The *instructor-regulated* (IRL) track encodes the slow-changing boundary conditions an instructor sets; the *self-regulated* (SRL) track carries the fast-changing per-student state that the system maintains on the learner’s behalf. Most of these components are inherited from or are standard extensions of our prior instructor-regulated system, SAGE [8]; the new mechanism this paper contributes is the factored learner model in the SRL track, and we treat the remaining components as the context it operates in.

A. Instructor-Regulated Track (inherited)

The **instructor model** stores and enforces instructor-set rules: which topics are unlocked, assignment lifecycle, and access control. It determines which concepts are eligible for evidence collection at all—if an instructor disables a topic, its trackers receive no updates. The **domain model** represents the subject as a knowledge graph whose edges encode prerequisite relations, together with a vector store used to retrieve course material for grounded generation [17]. The prerequisite edges are what the gate later checks a student’s mastered set against. Both components are inherited from SAGE and define the curriculum boundaries within which student progression is regulated [2].

B. Self-Regulated Track

The **perception layer** processes every student message through a short pipeline before anything else runs: it clas-

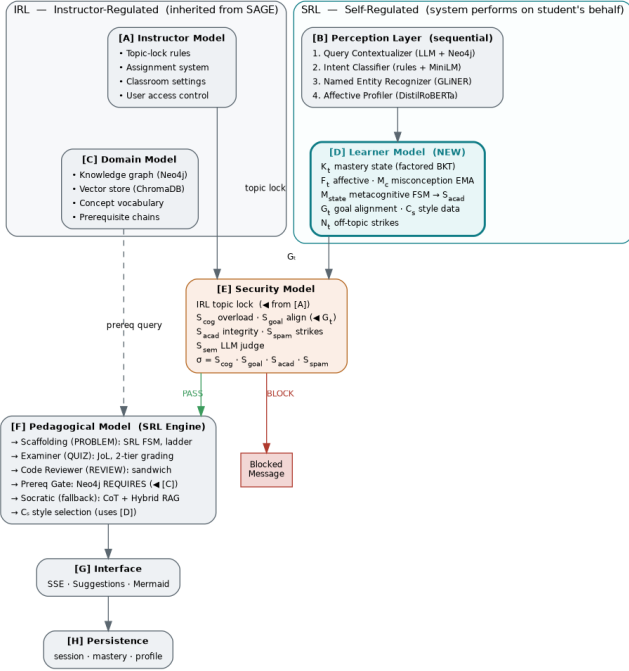


Fig. 2: System architecture. The instructor-regulated track (grey, inherited from SAGE) sets curriculum boundaries; the self-regulated track maintains per-student state. The factored learner model (D, highlighted) is the contribution; the gate, pedagogical engine, and infrastructure are inherited. All processing flows through the gate that consumes the mastered set K_t .

sifies the student’s intent, extracts which concept the interaction concerns, assigns the interaction to one of three evidence types (quiz, code attempt, or code review), and detects affective state [18]. This is the upstream computation that labels each interaction so it can be routed correctly. The unit of routing is the *cognitive level of the item*, not its delivery format. Modality is an imperfect proxy: a multiple-choice question asking a student to trace a loop’s output is quiz-format evidence of *applied* reasoning, and a fill-in-the-blank snippet is code-format evidence of *recall*. Each item in the authored question and exercise pools therefore carries an explicit cognitive-level tag, assigned at authoring time in the manner of a Q-matrix entry [?], and the router dispatches on the tag; modality serves as the default only for dynamically generated interactions, where format and level coincide by construction of the generation prompts. Robustness of the certification guarantee to routing error is quantified in Section IV-C.

The **learner model** is the core of this paper. For each concept it maintains three independent trackers—one per cognitive level—and updates exactly the one matching the perceived evidence type, certifying mastery only when all three are satisfied at once. This is the component that turns a stream of mixed interactions into a competency-certified record of mastered concepts; its formal construction, the routing rule, and the guarantee it satisfies are the subject of Section ?? . Surrounding the trackers, and feeding off the same interaction stream, are the supporting mecha-

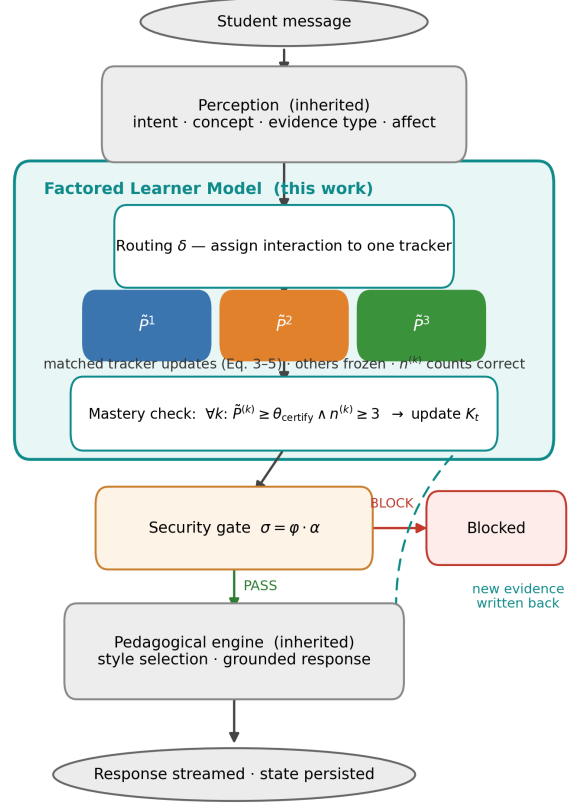


Fig. 3: Processing pipeline for a single interaction. The contribution (tinted) routes each interaction to one tracker, re-evaluates mastery, and updates K_t ; only requests passing the security gate reach the pedagogical engine, which writes new evidence back to the learner model.

nisms that make the model usable in a classroom: spaced-repetition scheduling for retention, intent-driven teaching-style selection with per-style feedback tracking, an exponentially weighted misconception monitor, and the affective signal from the perception layer. None of these resolve the Evidence Diversity Problem; all are needed for the learner model to run reliably at scale.

C. Convergence: Gate, Engine, and Persistence

The two tracks meet at the **security model**, a two-stage gate $\sigma = \varphi \cdot \alpha$ inherited from SAGE: a hard structural stage φ checks topic enablement, prerequisite satisfaction over the mastered set, content safety, and rate limits, and a soft stage α checks that the request aligns with the student’s stated goal. Only interactions that clear the gate reach the **pedagogical engine**, which selects a teaching style, generates the grounded response, and writes the resulting evidence back to the learner model. A streaming **interface** delivers responses and an instructor classroom view, and a **persistence** layer stores the learner state that must survive across sessions.

TABLE II: Subskill trackers and their initialization parameters. $\tilde{P}^{(k)}$ is the unconstrained BKT posterior in $[0, 1]$ used for all mastery decisions; the weight $\theta_{\max}^{(k)}$ scales only the displayed composite score $P^{(k)} = \theta_{\max}^{(k)} \tilde{P}^{(k)}$ and is never applied during inference. Non-zero priors follow Cromwell’s rule: a prior of exactly zero is an absorbing state of the Bayesian update, which no evidence can escape.

Subskill k (evidence)	$p_0^{(k)}$	$p_T^{(k)}$	$p_S^{(k)}$	$p_G^{(k)}$	$\theta_{\max}^{(k)}$
1 Declarative (quiz)	0.30	0.09	0.10	0.20	0.60
2 Procedural (micro)	0.05	0.09	0.15	0.10	0.25
3 Applied (code review)	0.01	0.09	0.20	0.05	0.10

D. Why the Tracks Are Separated

The split is deliberate. Instructor constraints are stable and change on the timescale of a course; student state changes with every interaction. Holding them in one model couples concerns that evolve at different rates and makes each harder to test and maintain. Separating them lets the instructor track be configured once per course while the learner track updates continuously, and it is precisely the learner track—the mastery signal—whose absence left the inherited gate acting on an asserted rather than an earned mastery set. The next section formalizes that signal.

IV. METHODOLOGY

We describe the learner model that produces a competency-certified record of mastered concepts, the guarantee it satisfies, and the supporting components that allow it to run inside an instructor-regulated tutor. Throughout, $\mathbf{1}[\cdot]$ denotes the indicator function.

A. Learner State

For a student u and concept c , the learner state at interaction t is

$$S_{u,c}(t) = (\tilde{P}_t^{(1)}, \tilde{P}_t^{(2)}, \tilde{P}_t^{(3)}, M_c(t), F_t, G_t), \quad (1)$$

where $\tilde{P}_t^{(k)} \in [0, 1]$ is the posterior mastery estimate for cognitive subskill k , $M_c(t)$ a misconception score, F_t a frustration level, and G_t a goal-alignment score. The first three entries are the heart of the model; the remainder support it and are defined in Section IV-E. For dashboard display we also report a Bloom-weighted composite $S_c(t) = \sum_{k=1}^3 \theta_{\max}^{(k)} \tilde{P}_t^{(k)} \in [0, 0.95]$, defined in Section IV-B; it is a display summary, not a decision variable.

B. Factored Multi-Skill Knowledge Tracing

Rather than one latent state per concept, the model maintains three independent Bayesian Knowledge Tracing trackers [3], one per Bloom-aligned cognitive level [9]: declarative (recall), procedural (code-writing), and applied (code-reasoning). Each tracker k has its own prior $p_0^{(k)}$, learning rate $p_T^{(k)}$, slip $p_S^{(k)}$, guess $p_G^{(k)}$, and a display weight $\theta_{\max}^{(k)}$ (Table II).

The priors encode how much competence a student plausibly brings to a concept before any practice: prior classroom

exposure makes some recall likely ($p_0^{(1)} = 0.30$), while procedural and applied competence require deliberate practice and are initialized near—but, by Cromwell’s rule, not at—zero ($p_0^{(2)} = 0.05$, $p_0^{(3)} = 0.01$). Slip and guess parameters reflect the response formats: multiple-choice quizzes admit substantial guessing ($p_G^{(1)} = 0.20$) but few slips, whereas free-form code review is nearly guess-proof ($p_G^{(3)} = 0.05$) but error-prone under genuine competence ($p_S^{(3)} = 0.20$). All values satisfy the standard non-degeneracy constraint $p_S^{(k)} + p_G^{(k)} < 1$ [?]. These are initialization defaults to be refit per concept by expert-constrained maximum likelihood once pilot data accumulate; Section V-? reports a sensitivity analysis showing that the structural results below are stable under perturbation of every parameter.

Evidence-type routing. Each interaction produces a pair (x_t, e_t) , where $x_t \in \{0, 1\}$ is correctness and $e_t \in \{\text{quiz}, \text{micro}, \text{code}\}$ is the evidence type. A binary selector routes the observation to exactly one tracker,

$$\delta^{(k)}(e_t) = \mathbf{1}[k \text{ corresponds to } e_t], \quad (2)$$

with $\delta^{(1)}(\text{quiz}) = \delta^{(2)}(\text{micro}) = \delta^{(3)}(\text{code}) = 1$ and all other entries zero. When $\delta^{(k)}(e_t) = 0$, tracker k is frozen: $\tilde{P}_{t+1}^{(k)} = \tilde{P}_t^{(k)}$. This is the mechanical core of the model—quiz evidence cannot move the procedural or applied trackers.

Posterior update. When $\delta^{(k)}(e_t) = 1$, the standard BKT evidence update is applied. For a correct observation,

$$P' = \frac{\tilde{P}_t^{(k)}(1 - p_S^{(k)})}{\tilde{P}_t^{(k)}(1 - p_S^{(k)}) + (1 - \tilde{P}_t^{(k)})p_G^{(k)}}, \quad (3)$$

and for an incorrect one,

$$P' = \frac{\tilde{P}_t^{(k)}p_S^{(k)}}{\tilde{P}_t^{(k)}p_S^{(k)} + (1 - \tilde{P}_t^{(k)})(1 - p_G^{(k)})}. \quad (4)$$

A learning transition is then applied *only on correct responses*:

$$\tilde{P}_{t+1}^{(k)} = \begin{cases} P' + (1 - P')p_T^{(k)}, & x_t = 1 \text{ (correct)}, \\ P', & x_t = 0 \text{ (incorrect)}. \end{cases} \quad (5)$$

Classical BKT applies the transition at every practice opportunity regardless of correctness [?]. We deviate deliberately: at a low-prior tier, an unconditional transition of $p_T = 0.09$ outweighs the Bayesian penalty for an error, so a wrong answer would *raise* the posterior—and, iterated, would let a persistently incorrect student drift upward until two lucky guesses certify the tier. Conditioning the transition on correctness removes this pathology while leaving the all-correct trajectory unchanged. Learning from errors is not discarded; in this architecture it is realized through the tutor’s feedback loop, which elicits a subsequent observation that updates the posterior through the ordinary evidence path. Because $\tilde{P}^{(k)}$ is an unconstrained posterior in $[0, 1]$, the

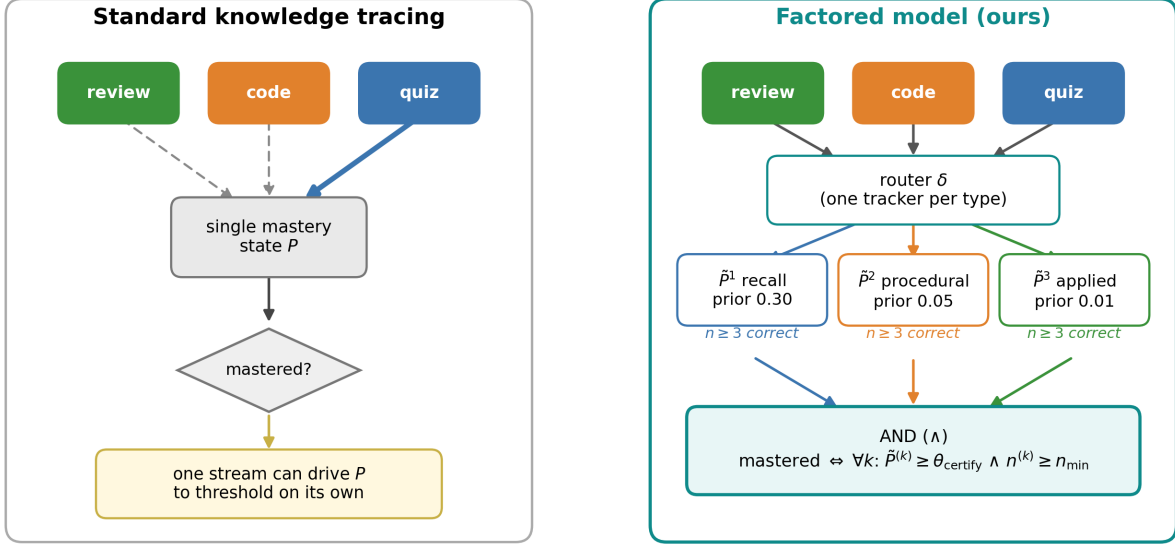


Fig. 4: Single-stream versus factored mastery. Standard knowledge tracing pools all evidence into one estimate (left), so a single stream can drive it to threshold; the factored model routes each evidence type to its own capped tracker and certifies mastery only when all three clear threshold (right).

weight $\theta_{\max}^{(k)}$ scales only the composite display score $P^{(k)} = \theta_{\max}^{(k)} \tilde{P}^{(k)}$ and plays no role in inference or certification; the composite $S_c = \sum_k \theta_{\max}^{(k)} \tilde{P}^{(k)} \in [0, 0.95]$ is a Bloom-weighted dashboard summary, not a decision variable. The three trackers are modeled as conditionally independent for tractability, following classical BKT [3]; this is a state-estimation convenience, not a claim that the competencies are cognitively independent.

C. Mastery Decision and the Evidence Diversity Guarantee

A concept is certified as mastered only when every tier simultaneously satisfies a posterior condition and an evidence-count condition:

$$m_c = 1 \iff \forall k \in \{1, 2, 3\} : \tilde{P}_t^{(k)} \geq \theta_{\text{certify}} \wedge n_t^{(k)} \geq n_{\min}, \quad (6)$$

with $\theta_{\text{certify}} = 0.95$ following the classical BKT mastery convention [?] and $n_{\min} = 3$. The counter $n_t^{(k)}$ records *correct* type- k observations only, matching the quantity over which Corollary 3 is stated. Certification is hysteretic: once certified, a concept remains in the mastered set until some tier's posterior falls below $\theta_{\text{decertify}} = 0.75$, preventing oscillation at the certification boundary. Both thresholds are initialization defaults, replaced at runtime by the calibrator of Section IV-D under constraints (Proposition 5) that preserve every result below. A concept enters the mastered set K_t exactly when $m_c = 1$; this is the set the instructor-regulated gate consumes.

Lemma 1 (Minimum evidence). *Under updates (3)–(5) with the parameters of Table II, no tier reaches θ_{certify} in fewer than three interactions, at least three of which must be correct.*

Proof. The correct- and incorrect-observation update maps are both monotone in $\tilde{P}$, and the former dominates the latter pointwise whenever $p_S^{(k)} + p_G^{(k)} < 1$, which every tier satisfies; hence the all-correct trajectory upper-bounds any history of equal length. From the priors $(0.30, 0.05, 0.01)$, two correct observations yield $(0.917, 0.849, 0.832) < 0.95$, while three yield $(0.982, 0.981, 0.989)$. The count condition in (6) supplies the correctness requirement. $\square$

Theorem 2 (Evidence Diversity Guarantee). *Let each tier's no-evidence dynamics never raise its posterior above its prior $p_0^{(k)}$. Under routing (2) and criterion (6) with $\theta_{\text{certify}} > \max_k p_0^{(k)}$, $m_c = 1$ implies the history contains at least $n_{\min}$ correct observations of every evidence type. Mastery cannot be certified from any single evidence type, however long the history.*

Proof. If tier k' never receives type- $e^{(k')}$ evidence, then $\delta^{(k')}(e_\tau) = 0$ for all τ , and tier k' evolves only under its no-evidence dynamics. Both such dynamics in this paper satisfy the hypothesis: the frozen-tracker rule holds the posterior at $p_0^{(k')}$, and the forgetting decay of Section IV-E has $p_0^{(k')}$ as its fixed point. Thus $\tilde{P}_t^{(k')} = p_0^{(k')} < \theta_{\text{certify}}$, contradicting $m_c = 1$. The count condition then requires $n_{\min}$ correct routed observations per type. $\square$

The guarantee depends only on routing exclusivity and $\theta_{\text{certify}} > p_0^{(k)}$, so it is a property of the architecture rather than of the calibrated parameter values.

Corollary 3 (False-certification bounds). *For a non-master answering tier k at guess rate $p_G^{(k)}$: (i) over a fixed battery of $n_{\min} = 3$ questions, the probability of certifying tier k is at most $(p_G^{(k)})^3$ — 0.80%, 0.10%, and 0.013% for*

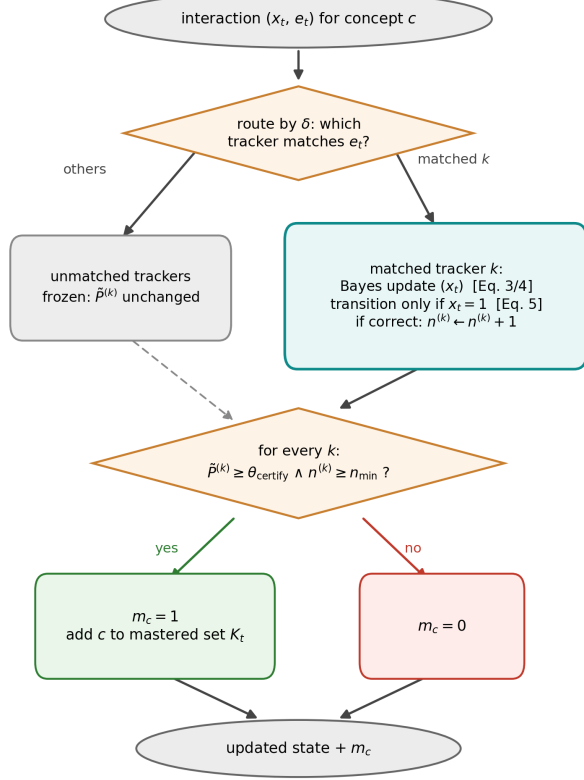


Fig. 5: Per-interaction update. Routing δ selects the one tracker matching the evidence type; the matched tracker applies the BKT update and ceiling clip while the others are frozen; mastery is the conjunction of three threshold tests, and a satisfied concept enters the mastered set K_t .

the declarative, procedural, and applied tiers—and at most 10^{-9} for a full concept; (ii) over an open-ended stream of length T , a union bound over three-correct windows gives $\Pr[\text{certify by } T] \leq (T - 2)(p_G^{(k)})^3$; exact hitting probabilities are reported in Section ??.

Remark 4. The guarantee assumes correct evidence typing by the perception layer. Under independent misclassification with rate ε , spuriously certifying a tier requires $n_{\min}$ independent routing errors, bounding the probability by $\binom{T}{n_{\min}} \varepsilon^{n_{\min}}$; we revisit this in the threats to validity (Section V).

D. Adaptive Threshold Calibration

The certification threshold in (6) is initialized at the literature-backed $\theta_{\text{certify}} = 0.95$ [?] but is not fixed. For each (concept, tier) pair the system maintains a sliding window of the $N = 500$ most recent posterior values. Once 50 observations accumulate, and every 50 thereafter, a two-component Gaussian mixture is fit to the window by EM, and a data-driven threshold is taken as the 10th percentile of the higher (mastered) component,

$$\theta_{\text{GMM}} = \mu_m - z_{0.10} \sigma_m, \quad z_{0.10} = 1.2816, \quad (7)$$

then blended with the prior in proportion to how cleanly the components separate:

$$\theta_{\text{new}} = (1 - w) \theta_{\text{prior}} + w \theta_{\text{GMM}}, \quad w = \frac{F}{1 + F}, \quad (8)$$

where $F = (\mu_m - \mu_o)^2 / (\sigma_m^2 + \sigma_o^2)$ is the Fisher discriminant ratio between the mastered and non-mastered components. Well-separated clusters drive $w \rightarrow 1$ and the data dominate; overlapping clusters or sparse data drive $w \rightarrow 0$ and the system reverts to the prior, a warm start that is down-weighted as evidence grows but never discarded. Below 50 observations the prior is used directly.

The calibrated value is clamped, per tier, to

$$\theta_{\text{certify}}^{(k)} \in \left[\max\{\tilde{P}_{2C}^{(k)}, \theta_{\text{decertify}}\} + \delta, 0.99 \right], \quad (9)$$

where $\tilde{P}_{2C}^{(k)}$ is the posterior after two consecutive correct responses from the tier prior (0.917, 0.849, 0.832; Lemma 1) and $\delta = 0.01$. The lower clamp leaves the calibrator genuine room below the 0.95 default while the upper clamp prevents a tightly concentrated mastered cluster—the expected regime, since uncensored posteriors accumulate near 0.98–0.99—from driving the threshold toward 1 and making certification unattainable. Threshold trajectories are logged per (concept, tier) so that certification decisions remain auditable as thresholds evolve.

Proposition 5 (Calibration invariance). *For every value the calibrator can produce under (9): (i) $\theta_{\text{certify}}^{(k)} > p_0^{(k)}$, so Theorem 2 holds; (ii) $\theta_{\text{certify}}^{(k)} > \tilde{P}_{2C}^{(k)}$, so no two-interaction history certifies and Lemma 1 holds; (iii) $\theta_{\text{certify}}^{(k)} > \theta_{\text{decertify}}$, so the hysteresis band never inverts. The count condition $n^{(k)} \geq n_{\min}$ is independent of the threshold and binds for every calibrated value.*

Two limitations are worth stating. A Gaussian mixture is misspecified for data supported on $[0, 1]$; a Beta mixture is the theoretically correct replacement and, now that the posteriors are uncensored with interior distributions, a direct substitution planned for future work, with the Fisher-ratio weighting limiting the damage in the interim by down-weighting exactly the poorly separated fits where misspecification is most severe. And the window pools posterior trajectories across students, whose within-student observations are autocorrelated; the practical effect is an understated σ_m , which the clamp bounds, and per-student subsampling is a planned refinement.

E. Supporting Components

The following components do not address evidence diversity but make the model usable in a deployed tutor; each is a local optimizer of one term in the utility metric of Section IV-F.

Misconception tracking. A high-confidence error is the signal of a stable misconception rather than noise. For concept c ,

$$M_c(t+1) = \lambda M_c(t) + (1 - \lambda) \mathbf{1}[\text{conf}_t \geq \theta_{\text{conf}} \wedge x_t = 0], \quad (10)$$

with $\lambda = 0.7$ and a flag raised when $M_c \geq 0.25$; on a correct response the score simply decays. A single confident error raises the score to 0.30 and flags immediately; one correct response decays it to 0.21, below the threshold, clearing a fresh misconception. A misconception reinforced by repeated high-confidence errors accumulates a higher score and requires proportionally more correct responses to clear—capturing the greater persistence of deeply held incorrect beliefs.

Spaced repetition. Review intervals follow SM-2, with the quality score $q \in [0, 5]$ derived from both correctness and the student’s reported confidence (a Judgment of Learning), and a retention signal $R_t = \mathbf{1}[q \geq 3]$.

Forgetting and decay. Between sessions the posterior is subject to forgetting. For tier k with last interaction at t_0 and elapsed time Δt in days,

$$\tilde{P}^{(k)}(t) = \tilde{P}^{(k)}(t_0) e^{-\lambda^{(k)} \Delta t} + p_0^{(k)} (1 - e^{-\lambda^{(k)} \Delta t}), \quad (11)$$

an exponential relaxation whose fixed point is the tier prior, so an untouched tracker remains at its prior and Theorem 2 is unaffected. The rate $\lambda^{(k)}$ sets the half-life of the excess posterior $\tilde{P}^{(k)} - p_0^{(k)}$; initial half-lives of 14, 7, and 5 days for the declarative, procedural, and applied tiers reflect the finding that declarative knowledge decays more slowly than procedural and applied skill [19]. Decay and SM-2 are coupled rather than parallel: after each successful review, $\lambda_{i+1}^{(k)} = \lambda_i^{(k)} / \text{EF}_i$ with the SM-2 ease factor $\text{EF}_i \geq 1.3$, so each review lengthens the memory half-life—the spacing effect [20], [21]. The review interval is derived from the same model, as the time at which the posterior would decay to the decertification threshold:

$$I^{(k)} = -\frac{1}{\lambda^{(k)}} \ln \frac{\theta_{\text{decertify}} - p_0^{(k)}}{\tilde{P}^{(k)}(t_0) - p_0^{(k)}}. \quad (12)$$

A freshly certified declarative tier ($\tilde{P} = 0.982$), for example, is first scheduled for review after $I^{(1)} \approx 8.4$ days.

Knowledge credibility verification. A student may claim prior knowledge of a concept (“I know X—verify”). The system responds with a battery of $n_{\min} = 3$ independent questions from the concept’s declarative pool, each graded as one ordinary BKT update. The *declarative tier* certifies in this single session only if all three are correct, which from the tier prior yields a posterior of 0.982 and a count of $n^{(1)} = 3$. Concept-level certification is unaffected: the procedural and applied tiers still require their own evidence under Theorem 2, so a verified claim accelerates progress through recall material without reopening single-stream certification, and the fixed-battery bound of Corollary 3(i) applies unchanged.

Teaching-style selection. Thumbs-up and thumbs-down feedback is recorded as per-style win and total counts in the learner state. Style selection currently follows the classified intent of the interaction; an explicit exploration–exploitation policy over the intent-compatible styles (such as UCB1 on the recorded counts) is a planned extension.

Affect. An affective profiler maps the student’s emotional state to a frustration level F_t used to soften tone and pacing, so that frustration does not corrupt the quality of the collected evidence.

Security gate. Inherited from SAGE [22] and expressed here as $\sigma(x_t) = \varphi(x_t) \cdot \alpha(x_t)$, the gate combines a hard structural stage φ —enforcing topic enablement, prerequisite satisfaction over K_t , content safety, and rate limits—with a soft alignment stage $\alpha = \mathbf{1}[G_t \geq \theta_{\text{goal}}]$, where the goal-alignment score G_t is the rescaled cosine similarity between the current request and the student’s stated goal g_0 . Because φ multiplies α , a structural block cannot be overridden by alignment. Because forgetting decay can carry a posterior below θ_{certify} between sessions, the prerequisite check in φ is evaluated over the *ever-certified* set—a permanent record of concepts that have at some time satisfied criterion (6)—while the live posterior drives only the review scheduler. This prevents downstream topics from silently re-locking when a student is merely due for review; the instructor dashboard distinguishes *certified*, *current* from *certified*, *review due*.

Relation to self-regulated learning. The system performs much of a learner’s self-regulation on their behalf rather than building it: the trackers monitor the learning state, M_c catches confident errors, and the decay-coupled SM-2 scheduler manages review. The student supplies three genuine metacognitive signals—a goal statement g_0 , checked for alignment by α ; confidence ratings, which feed the quality score q , the misconception monitor, and the evidence updates; and explicit prior-knowledge claims, tested by the credibility battery.

F. System Utility Metric

The components are tied together by a single objective that states what the system is trying to achieve:

$$J = \sum_t [w_1 \Delta K_t + w_2 R_t - w_3 F_t - w_4 C_t], \quad (13)$$

where ΔK_t is the per-step mastery gain, R_t the retention signal, F_t the frustration cost, and C_t the instructional cost (higher when the tutor transfers more cognitive labor to itself, e.g. a full handout versus a scaffolded hint). We emphasize that J is not jointly optimized in the current system: each component optimizes its own term locally, and J serves as an explicit system-utility metric and an evaluation target rather than a loss minimized end-to-end. Joint optimization, for instance by reinforcement learning, is left to future work.

V. PERFORMANCE EVALUATION

Placeholder for evaluation results.

VI. CONCLUSION

Placeholder for conclusion.

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